CS3281 / CS5281

Concurrency Bugs

CS3281 / CS5281

Spring 2026

**Some lecture slides borrowed and adapted from Branden Ghena and Stephen Tarzia (Northwestern), Harsha Madhyastha (Michigan), Shivaram Venkataraman (Wisconsin), and UC Berkeley CS162*

Already Learned

- The need for synchronization
- Mutexes `(pthread_mutex_t)`
- Condition Variables `(pthread_cond_wait)`
- Semaphores `(sem_t)`
- Futexes

Today's Goals

- Common synchronization bugs
  - Deadlock
  - Livelock
- Methods to avoid, prevent, and recover in the presence of deadlock
- Touch on what concurrency looks like in other languages

Outline

- **Synchronization bugs**
- Deadlock
  - Solving Deadlocks
- Livelock
- Other languages

Common synchronization bugs

- Atomicity violation
  - Critical section is violated (due to missing lock)
- Order violation
  - Something happens sooner (or later) than expected
- Deadlock
  - Two threads wait indefinitely on each other
- Livelock (not that common in practice)
  - Two threads repeatedly block each other from proceeding and retry

Atomicity violation

- It's relatively easy to find and protect critical sections,
  - But often we forget to add locks around other uses of the shared data.
- Obvious critical section is here:
  - Two threads should not enter this at once
- But, we also have to make sure that *file* is not modified elsewhere.
- Even if this one-line *close* was atomic we have to make sure it doesn't run during the above critical section.

```
Main Thread
lock(lck);
if (file == NULL) {
  file = open("~/myfile.txt");
}
write(file, "hello file");
unlock(lck);

…
Some Other Thread
close(file);   // whoops!!
```

Order violation

- Code often requires a certain ordering of operations, especially:
  - Objects must be initialized before they're used
  - Objects cannot be freed while they are still in use

Parent Thread

```
file = open("file.dat");
thread_create(child_fcn);
// do some work
…
close(file);
```

Child Thread

```
child_fcn() {
  write(file, "hello");
}
```

Close must happen after *write*, but code does not enforce this ordering.

Why is this difficult?

- It seems like we can just add lots of locks and CVs to be safe, right?
  - Still tricky! Too many locks can cause deadlock – indefinite waiting.
- How about just one big lock?
  - (+) Cannot deadlock with one lock
  - (–) However, this would ***limit concurrency***
    - If every task requires the same lock, then unrelated tasks cannot proceed in parallel.
- Tradeoff: Programming headache vs. Performance

Locking granularity

- ***Coarse grained lock*:**
  - Use one (or a few) locks to protect all (or large chunks of) shared state
  - Linux kernel < version 2.6.39 used one “Big Kernel Lock”
  - Essentially only one thread (CPU core) could run kernel code
  - It’s simple but there is much contention for this lock, and concurrency is limited

- ***Fine grained locks*:**
  - Use many locks, each protecting small chunks of related shared state
  - Leads to more concurrency and better performance
  - However, there is greater risk of ***deadlock***

Outline

- Synchronization bugs
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  - Solving Deadlocks
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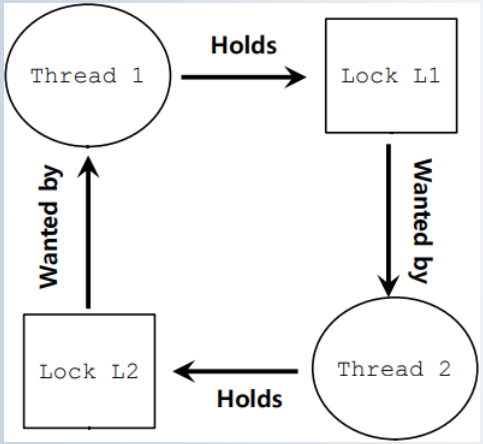
Deadlock

- Def: A process is *deadlocked* iff it is waiting for a condition that will never be true

- A concurrency bug arising usually when:
  - Processes 1 and 2 needs two resources (A and B) to proceed
  - Process 1 acquires A, waits for B
  - Process 2 acquires B, waits for A
  - Both will wait forever!

Deadlock Bugs

- Deadlock: A thread holds a lock and waits for another lock.
  - Happens when ***multiple*** shared resources are used.



Thread 1:
pthread_mutex_lock (L1);
context switch to Thread 2
pthread_mutex_lock (L2);

Thread 2;
pthread_mutex_lock (L2);

pthread_mutex_lock (L1);

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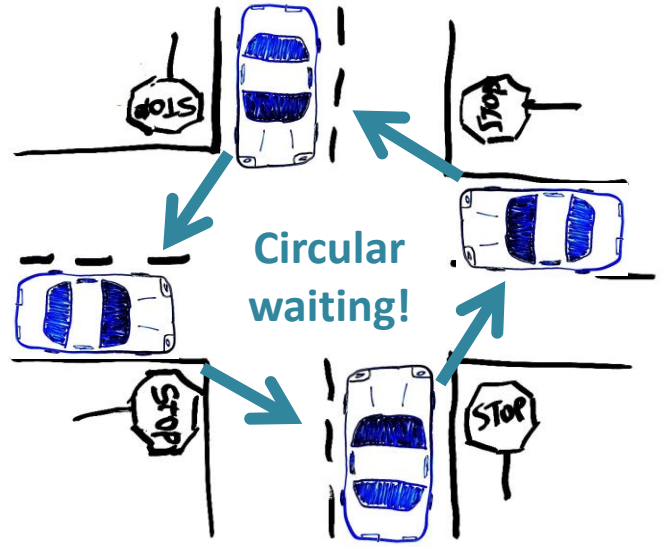
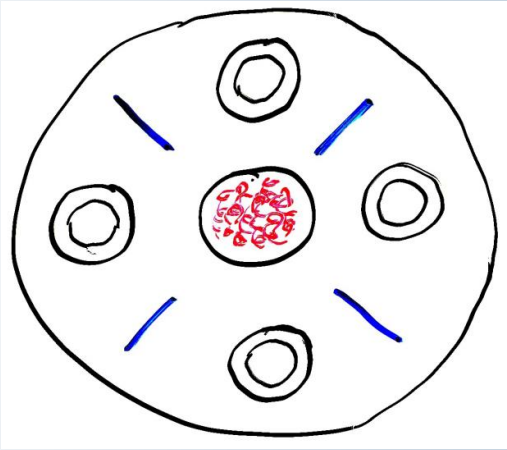
Simple example: four-way stop

- Traffic rules state that you must **yield to the car on your right** if you reach the intersection simultaneously.
- This rule usually works well.
- But there's a problem if four cars arrive simultaneously.



Dining philosophers

- A theoretical example of deadlock
- There are N philosophers sitting in a circle and N chopsticks
  - left and right of each philosopher
- Philosophers repeatedly run this loop:
  1. Think for some time
  2. Grab chopstick to left
  3. Grab chopstick to right
  4. Eat
  5. Replace chopsticks
- If they all grab the left chopstick simultaneously (step 2), they will deadlock and starve!



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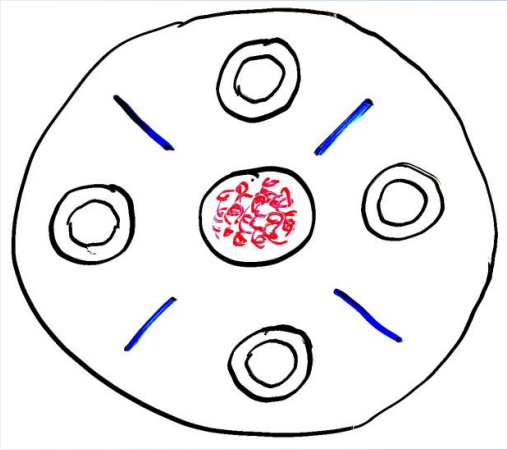
- A solution: one philosopher must grab right before left



Dining philosophers

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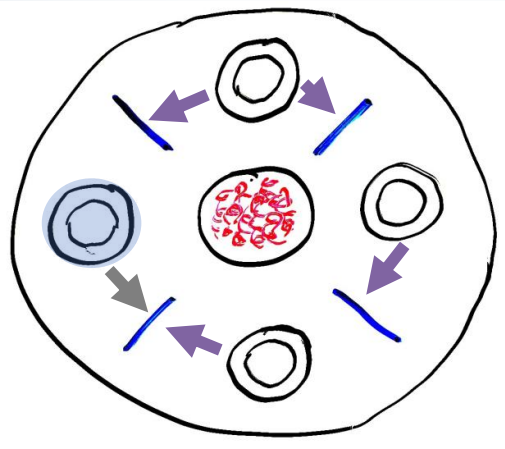
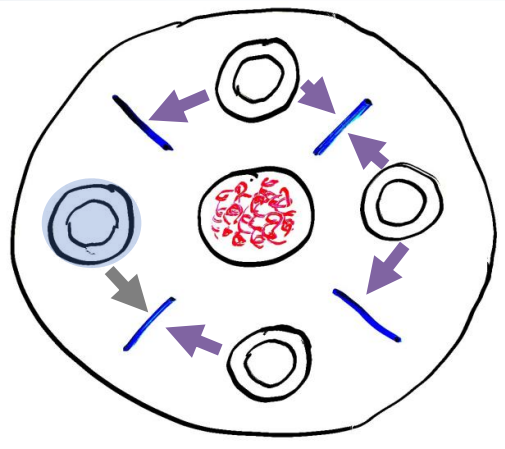


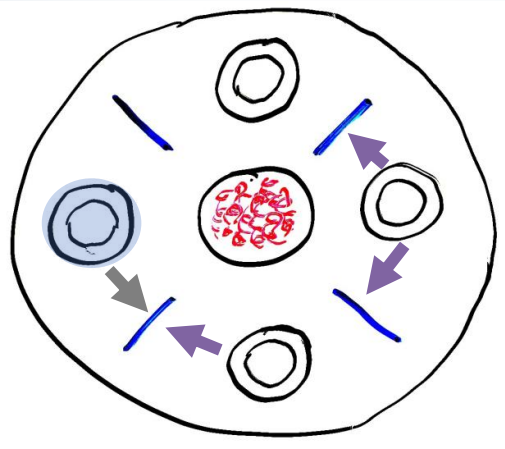
Dining philosophers

- A solution: one philosopher must grab right before left



Dining philosophers

- A solution: one philosopher must grab right before left
  - Adding an asymmetry will allow both resources to eventually be obtained



Deadlock with locks

Thread A

```
x.Acquire();
y.Acquire();
…
y.Release();
x.Release();
```

Thread B

```
y.Acquire();
x.Acquire();
…
x.Release();
y.Release();
```

- This is a Nondeterministic Deadlock
  - Whether it occurs depends on scheduling

No deadlock in the lucky case

Thread A

```
x.Acquire();
y.Acquire();

…
y.Release();
x.Release();
```

Thread B

```
y.Acquire();

x.Acquire();
…
x.Release();
y.Release();
```

Thread B waits until Thread A is finished

But deadlock can still occur

Thread A

```
x.Acquire();

y.Acquire();
```

Thread B

```
y.Acquire();

x.Acquire();
```

Thread A waits until y is available

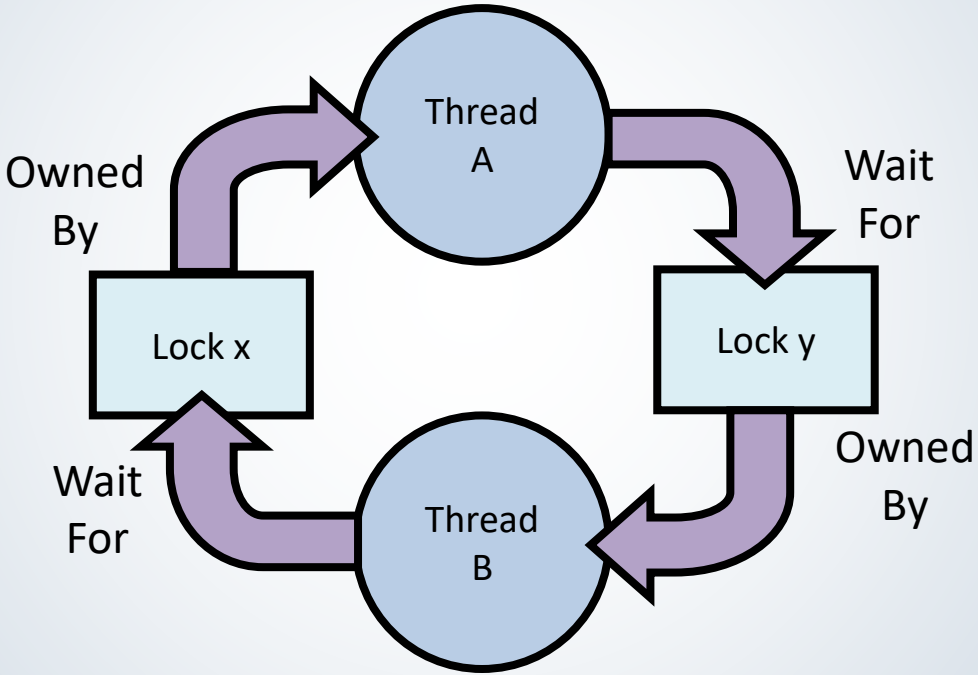
Thread B waits until x is available

--Unreachable--

```
…
y.Release();
x.Release();
```

```
…
x.Release();
y.Release();
```

Deadlocks involve *circular dependencies*



Deadlock can occur on any shared resource

- Example deadlock if the system only has 2 MB of memory

```
Thread A                Thread B
AllocateOrWait(1 MB)    AllocateOrWait(1 MB)
AllocateOrWait(1 MB)    AllocateOrWait(1 MB)
Free(1 MB)              Free(1 MB)
Free(1 MB)              Free(1 MB)
```

- Could deadlock on access to hardware as well

Interrupts can cause deadlocks too

```
Thread A
acquire()
                Interrupt Handler
…               acquire()
```

Deadlock

```
release()
```

- Thread cannot continue until the interrupt is finished
- Interrupt cannot finish until the thread continues

Outline

- Synchronization bugs
- **Deadlock**
  - **Solving Deadlocks**
- Livelock
- Other languages

How Should a System Deal With Deadlock?

- Three different approaches:

1. **Deadlock avoidance:** dynamically delay resource requests so deadlock doesn't happen

2. Deadlock prevention: write your code in a way that it isn't prone to deadlock

3. Deadlock recovery: let deadlock happen, and then figure out how to recover from it

Deadlock avoidance

- Idea: When a thread requests a resource, OS checks if it would result in an unsafe state that could lead to deadlock
  - If not, grant the resource
  - If so, wait until other threads release resources

Thread A	**Thread B**
`x.Acquire();`	
	`y.Acquire();`
	`x.Acquire();`
`y.Acquire();`	…
…	`x.Release();`
`y.Release();`	`y.Release();`
`x.Release();`	

Must stop acquire here to prevent unsafe state

Banker's Algorithm for avoiding deadlock

- Each thread states maximum resource needs in advance
- OS allows a particular thread to claim a resource if
  - (available resources - requested) >= min(remaining need of any thread)

- For Dining Philosophers, a request for a chopstick is allowed if:
  1. Not the last chopstick
  2. Or is the last chopstick but a philosopher will have two afterwards

How Should a System Deal With Deadlock?

- Three different approaches:

1. Deadlock avoidance: dynamically delay resource requests so deadlock doesn't happen

2. **Deadlock prevention:** write your code in a way that it isn't prone to deadlock

3. Deadlock recovery: let deadlock happen, and then figure out how to recover from it

Deadlock Bugs

- Conditions for deadlocks
  1. Mutual exclusion: A resource can be held by one thread at a time
  2. Hold-and-wait: Thread holds the lock and waits to acquire an additional lock
  3. No preemption: Lock that is held cannot be taken away from the thread
  4. Circular wait: *hold-and-wait* for a circular chain of threads

- Can eliminate deadlock by eliminating ***any*** one of these conditions

1. Do not have mutual exclusion

- Lockfree/waitfree data structures

```
void* mythread(void* arg) {
  for (int i=0; i<LOOPS; i++) {
    pthread_mutex_lock(&lock);
    counter++;
    pthread_mutex_unlock(&lock);
  }
  return NULL;
}
```

```
void* mythread(void* arg) {
  for (int i=0; i<LOOPS; i++) {
    atomic_fetch_and_add(
        &counter, 1);
  }
  return NULL;
}
```

2. Avoid hold and wait with trylock()

- We can avoid deadlock if we release the first lock after noticing that the second lock is unavailable.
- ***Trylock()*** tries to acquire a lock, but returns a failure code instead of waiting if the lock is taken:
- This code *cannot deadlock*, even if another thread does the same with L2 first, then L1.

```
top:
  lock(L1);
  if (trylock(L2) == -1) {
    unlock(L1);
    goto top;
  }
```

- However, it can lead to *livelock*… we'll come back to this

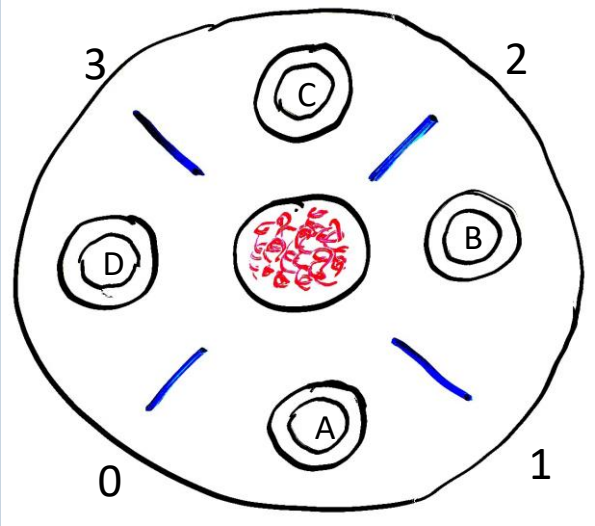
3. No preemption

- The OS *could* take away the lock from a blocked thread and give it back before the thread resumes
  - This sounds pretty complicated to get right
- Non-lock resources are easier here
  - Temporarily take away memory from a thread by swapping it to disk

4. Avoiding Circular Wait

- This is the most practical way to avoid deadlock.
- The simplest solution is to always acquire locks in the same order.
  - If you hold lock X and are waiting for lock Y,
  - Then holder of Y cannot be waiting on you,
  - Because they would have already acquired X before acquiring Y.
- However, in practice it can be difficult to know when locks will be acquired because they can be buried in subroutines.

Ordered locking for dining philosophers



- The chopsticks are shared resources, like locks
- If we require the **lower-numbered chopstick to be grabbed first**, this eliminates circular waiting.
  - Philosophers A, B, C grab *left then right*.
  - However, philosopher D will grab *right then left*.
  - If everyone tries to start at once, A & D race to grab chopstick 0 first, and the winner eats first.
  - While one is waiting to grab its first chopstick a neighbor will be able to grab two chopsticks.

Check your understanding

- In what order must Thread B acquire the three locks to avoid deadlock?

Thread A

```
y.Acquire();
x.Acquire();
z.Acquire();
…
z.Release();
x.Release();
y.Release();
```

Thread B

```
???
```

Check your understanding

- In what order must Thread B acquire the three locks to avoid deadlock?
  - The same order!! (at least y first, for the two-thread case)

Thread A	**Thread B**
`y.Acquire();`	`y.Acquire();`
`x.Acquire();`	`x.Acquire();`
`z.Acquire();`	`z.Acquire();`
…	…
`z.Release();`	`z.Release();`
`x.Release();`	`x.Release();`
`y.Release();`	`y.Release();`

How Should a System Deal With Deadlock?

- Three different approaches:

1. Deadlock avoidance: dynamically delay resource requests so deadlock doesn't happen

2. Deadlock prevention: write your code in a way that it isn't prone to deadlock

3. **Deadlock recovery:** let deadlock happen, and then figure out how to recover from it

Deadlock Recovery: how to deal with a deadlock?

- Terminate thread, force it to give up resources
  - Dining Philosophers Example: Remove a dining philosopher
  - In `AllocateOrWait` example, OS kills a process to free up some memory
  - Not always possible—killing a thread holding a lock leaves world inconsistent

- Roll back actions of deadlocked threads
  - Common techniques in databases (transactions)
  - Of course, if you restart in exactly the same way, you may enter deadlock again

Modern OS approach to deadlocks

- Make sure the *system* isn't involved in any deadlock
  - Hopefully by prevention
  - Generally, be very careful about this stuff in the kernel
- Ignore deadlock in applications ("Ostrich Algorithm")
  - User can just restart them anyways

Break + Check your understanding

- Is there a possibility of deadlock?
  - If so, how could we fix it?

Thread A

```
usb.Acquire();
webcam.Acquire();
…
webcam.Release();
usb.Release();
```

Thread B

```
printer.Acquire();
usb.Acquire();
…
usb.Release();
printer.Release();
```

Thread C

```
webcam.Acquire();
printer.Acquire();
…
printer.Release();
webcam.Release();
```

Break + Check your understanding

- Is there a possibility of deadlock? **Yes**
  - If so, how could we fix it? **One solution: Global ordering of resources**
    - Example: usb, then webcams, then printers always in that order

Thread A

```
usb.Acquire();
webcam.Acquire();
…
webcam.Release();
usb.Release();
```

Thread B

~~printer.Acquire();~~
~~usb.Acquire();~~
usb.Acquire();
printer.Acquire();
…
~~usb.Release();~~
~~printer.Release();~~
printer.Release();
usb.Release();

Thread C

```
webcam.Acquire();
printer.Acquire();
…
printer.Release();
webcam.Release();
```

Break + Check your understanding

- Is there a possibility of deadlock? **Yes**
  - If so, how could we fix it? **One big lock still works too!**

Thread A

```
lock.acquire();
usb.Acquire();
webcam.Acquire();
…
webcam.Release();
usb.Release();
lock.release();
```

Thread B

```
lock.acquire();
printer.Acquire();
usb.Acquire();
…
usb.Release();
printer.Release();
lock.release();
```

Thread C

```
lock.acquire();
webcam.Acquire();
printer.Acquire();
…
printer.Release();
webcam.Release();
lock.release();
```

Outline

- Synchronization bugs
- Deadlock
  - Solving Deadlocks
- **Livelock**
- Other languages

Livelock

- Implementing trylock to avoid deadlocks (break hold and wait)
- Use pthread_mutex_trylock()
  - Acquire lock if it is available or return an error code (lock is already held)
  - Preemption: release the ownership of a lock

```
top:
  lock(L1);
  if (trylock(L2) == -1) {
    unlock(L1);
    goto top;
  }
```

- Livelock
  - Another thread holds L2 and attempts to acquire L1.
  - Both threads execute their code blocks at the same time.

Livelock while avoiding deadlock

```
// thread 1
getLocks12(lock1, lock2) {
  lock1.acquire();
  while (lock2.locked()) {
    // attempt to step aside
    // for the other thread
    lock1.release();
    wait();
    lock1.acquire();
  }
  lock2.acquire();
}
```

```
// thread 2
getLocks21(lock1, lock2) {
  lock2.acquire();
  while (lock1.locked()) {
    // attempt to step aside
    // for the other thread
    lock2.release();
    wait();
    lock2.acquire();
  }
  lock1.acquire();
}
```

Avoiding hold and wait could lead to livelock

- Avoiding hold and wait can *livelock*
  - Two threads *could* get stuck in this loop forever
  - Unlikely to occur for any length in personal computing setting
  - Very possibly stuck forever (or at least extended periods) in a constrained computing setting
    - Example: embedded system with known tasks at the start

```
top:
  lock(L1);
  if (trylock(L2) == -1) {
    unlock(L1);
    goto top;
  }
```

Livelock in agents

- Livelock is more common in agent-based programs
  - All of agent's options lead to a lack of forward progress

- One example: video games
  - The character can still move and take actions
  - But cannot complete the level

Livelock versus Deadlock

```
top:
  lock(L1);
  if (trylock(L2) == -1) {
    unlock(L1);
    goto top;
  }
```

- Livelock is a condition where two threads repeatedly take action, but still don't make progress.
- Differs from deadlock because deadlock is always permanent.
- Livelock involves retries that ***may*** lead to progress, but there is no ***guarantee*** of progress.
- Any randomness in the timing of retries will fix livelock.
- In practice, livelock is a much less serious concern than deadlock.

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- Livelock
- **Other languages (extra)**

Javascript

- Javascript (in browsers) is strictly single-threaded
  - Therefore, no data races!

- A Javascript function will never be interrupted unless it makes an asynchronous call

```
console.log("1");
setTimeout(function(){console.log("2");},0);
console.log("3");
setTimeout(function(){console.log("4");},1000);
```

  - Will always output: **1 3 2 4** in that order
    - Even timers only trigger whenever the current code is finished

Python

- Provides all the same primitives we discussed! https://docs.python.org/3/library/concurrency.html

threading — Thread-based parallelism
- Thread-Local Data
- Thread Objects
- Lock Objects
- RLock Objects
- Condition Objects
- Semaphore Objects
  - Semaphore Example
- Event Objects
- Timer Objects
- Barrier Objects

And some nicer things

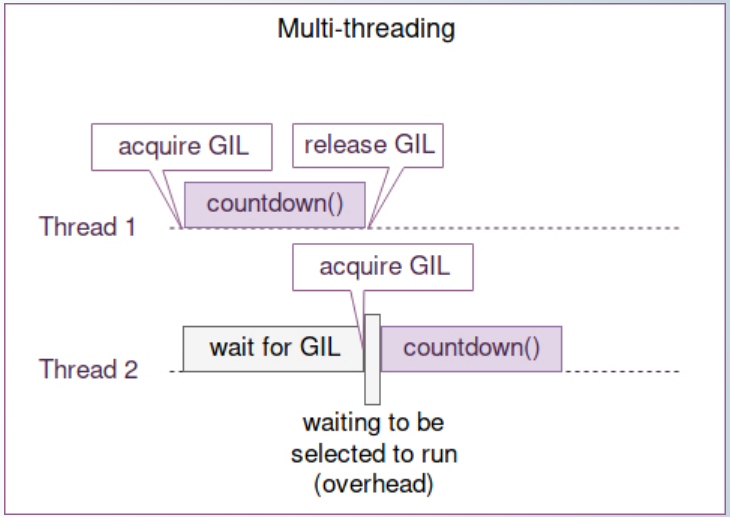
```
with some_lock:
  # do something…
```

Is equivalent to

```
some_lock.acquire()
try:
  # do something…
finally:
 some_lock.release()
```

Python threads are concurrent but not parallel

- Python uses one big lock technique for thread safety
  - Global Interpreter Lock (GIL)
  - Threads that are I/O bound still get a performance boost
  - Threads that are CPU bound do not increase performance



- *Multiprocessing* library does employ parallelism by spawning entirely new processes
  - Each with their own python interpreter

https://hackernoon.com/concurrent-programming-in-python-is-not-what-you-think-it-is-b6439c3f3e6a

Java

- Java has ***synchronized*** keyword for surrounding critical sections
- Automatically releases the lock when exiting early:

```
public class Counter {
    int mTotal = 0;

    public synchronized void addOne() {
        int val = mTotal;
        val++;
        mTotal = val;
    }

    public void addOneVersion2() throws Exception {
        int val;
        synchronized(this) {
            val = mTotal;
            val++;
            if (val == Integer.MAX_VALUE) {
                throw new Exception("value is too large");
            }
            mTotal = val;
        }
        System.out.println("new value is " + val);
    }
}
```

- Similar to
  - Python: "`with self.lock:`"
  - Objective-C: "`@synchronized`"

Rust

- Rust's opinion on sharing memory is amusingly to refer to Go's opinion

> *Do not communicate by sharing memory; instead, share memory by communicating.*
>
> --*Effective Go*

- Rust has a strong concept of ownership
  - A writeable (mutable) reference to an object can only be held in one place
  - Once an object is passed to another thread, the passer no longer has access
  - Solves many concurrency issues due to lack of shared memory

- Rust locks have lifetimes enforced by the compiler
  - Lock goes out-of-scope at the end of the function, relocking automatically

https://blog.rust-lang.org/2015/04/10/Fearless-Concurrency.html

Advice for the future

- Be aware of issues when writing multithreaded code
- Use thread-safe data structures when possible
  - In languages that provide them…
- Map your problem onto a classical concurrency problem
  - Producer/Consumer
  - Readers/Writers
- One big lock for *correctness* isn't the worst idea ever
  - But with some care (possibly a lot of care) we can do better

Examples

Deadlock Bugs

- Prevention for: Mutual Exclusion
  - Lock-free data structures
- Create atomic functions based on hardware instructions

```
int CompareAndSwap(int *address, int expected, int new){
    if(*address == expected){
            *address = new;
            return 1; // success
    }
    return 0;
}
```

```
void AtomicIncrement(int *value, int amount){
    do{
            int old = *value;
    }while( CompareAndSwap(value, old, old+amount)==0);
}
```

Deadlock Bugs

- List insertion
- Race condition occurs if called by multiple threads

```
void insert(int value){
    node_t * n = malloc(sizeof(node_t));
    assert( n != NULL );
    n->value = value ;
    n->next  = head;
    head     = n;
}
```

interrupt

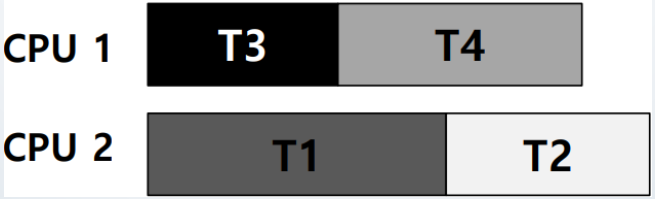
```
void insert(int value) {
    node_t *n = malloc(sizeof(node_t));
    assert(n != NULL);
    n->value = value;
    do {
            n->next = head;
    } while (CompareAndSwap(&head, n->next, n) == 0);
}
```

Deadlock Bugs

- Avoid deadlock
- Find out what locks are acquired by what threads

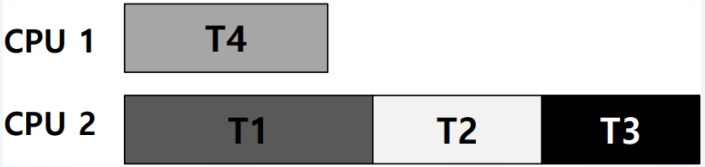
	T1	T2	T3	T4
L1	yes	yes	no	no
L2	yes	yes	yes	no

- Scheduler does not run T1 and T2 at the same time.
- T3 grabs only one lock. It can run with T1.



Deadlock Bugs

	T1	T2	T3	T4
L1	yes	yes	yes	no
L2	yes	yes	yes	no



- Decrease concurrency
- Trade-off between performance and deadlock avoidance
- It's impractical to gain a prior knowledge of lock acquisition for most applications.

Deadlocking with Semaphores

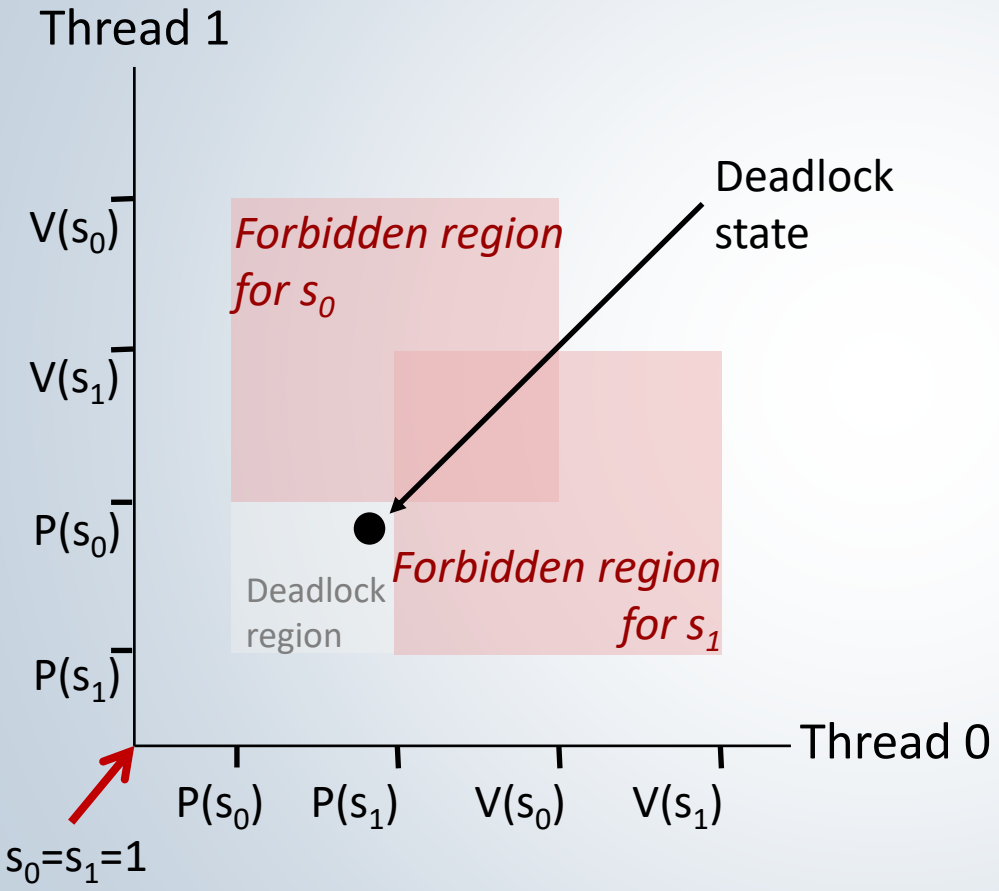
```
int main()
{
   pthread_t tid[2];
   Sem_init(&mutex[0], 0, 1);  /* mutex[0] = 1 */
   Sem_init(&mutex[1], 0, 1);  /* mutex[1] = 1 */
   Pthread_create(&tid[0], NULL, count, (void*) 0);
   Pthread_create(&tid[1], NULL, count, (void*) 1);
   Pthread_join(tid[0], NULL);
   Pthread_join(tid[1], NULL);
   printf("cnt=%d\n", cnt);
   exit(0);
}
```

```
void *count(void *vargp)
{
   int i;
   int id = (int) vargp;
   for (i = 0; i < NITERS; i++) {
            P(&mutex[id]); P(&mutex[1-id]);
            cnt++;
            V(&mutex[id]); V(&mutex[1-id]);
   }
   return NULL;
}
```

Tid[0]:
$P(s_0)$;
$P(s_1)$;
cnt++;
$V(s_0)$;
$V(s_1)$;

Tid[1]:
$P(s_1)$;
$P(s_0)$;
cnt++;
$V(s_1)$;
$V(s_0)$;

Deadlock Visualization in a Progress Graph



Locking introduces the potential for *deadlock:* waiting for a condition that will never be true

Any trajectory that enters the *deadlock region* will eventually reach the *deadlock state*, waiting for either s_0 or s_1 to become nonzero

Other trajectories luck out and skirt the deadlock region

Unfortunate fact: deadlock is often nondeterministic (race)

Avoiding Deadlock

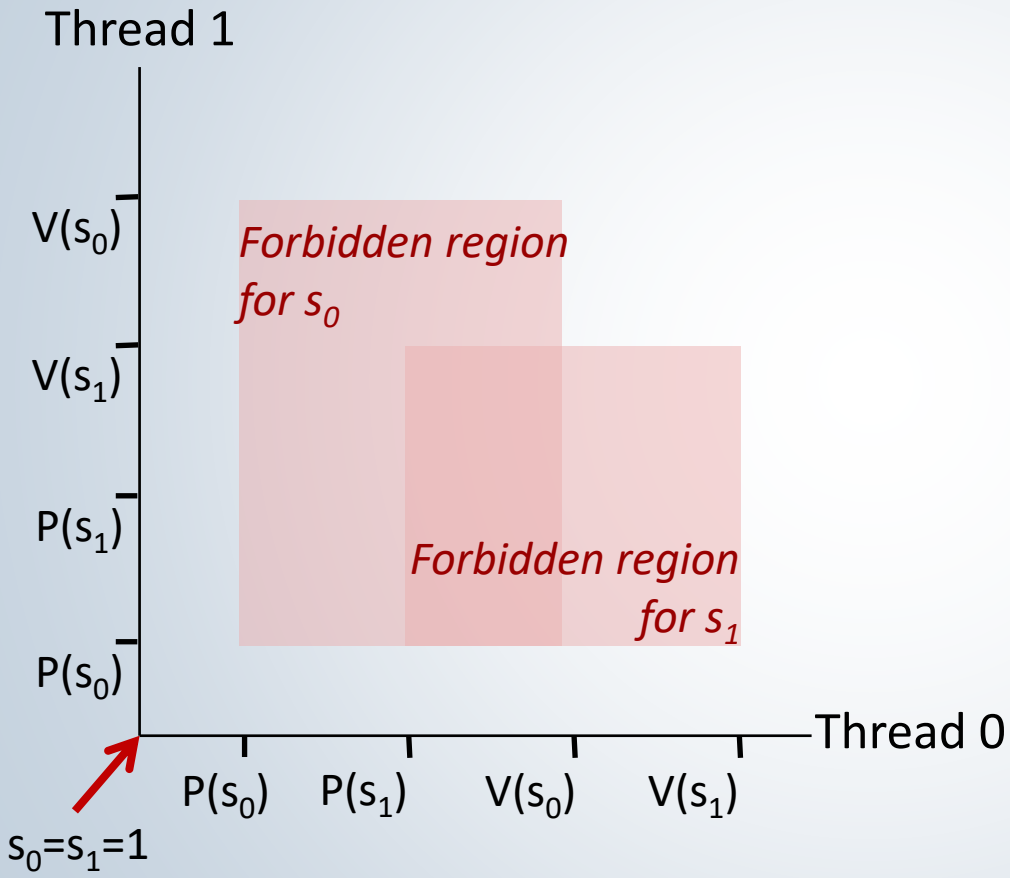
```
int main()
{
    pthread_t tid[2];
    Sem_init(&mutex[0], 0, 1);  /* mutex[0] = 1 */
    Sem_init(&mutex[1], 0, 1);  /* mutex[1] = 1 */
    Pthread_create(&tid[0], NULL, count, (void*) 0);
    Pthread_create(&tid[1], NULL, count, (void*) 1);
    Pthread_join(tid[0], NULL);
    Pthread_join(tid[1], NULL);
    printf("cnt=%d\n", cnt);
    exit(0);
}
```

```
void *count(void *vargp)
{
    int i;
    int id = (int) vargp;
    for (i = 0; i < NITERS; i++) {
        P(&mutex[0]); P(&mutex[1]);
     cnt++;
     V(&mutex[id]); V(&mutex[1-id]);
    }
    return NULL;
}
```

Tid[0]:
P(s0);
P(s1);
cnt++;
V(s0);
V(s1);

Tid[1]:
P(s0);
P(s1);
cnt++;
V(s1);
V(s0);

Avoiding Deadlock in a Progress Graph



No way for trajectory to get stuck

Processes acquire locks in same order

Order in which locks released immaterial

Summary

- Common solutions for deadlocks
  - Prevent deadlock
    - Order lock acquisition
    - Acquire lock atomically
    - Release lock voluntarily
    - Build lock-free atomic operations
  - Avoid deadlock
    - Schedule threads with global knowledge of lock acquisitions